

Lecture 4 - Discrete Random Variables

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Lecture 3 described probabilities of events and showed how to combine and condition them. We now use random variables to assign numerical values to outcomes and probability mass functions to describe how probability is distributed across those values.

This lecture develops discrete random variables, common discrete distributions, independence and conditional independence, expectation, and variance. It also distinguishes the probability of data under a fixed model from the likelihood of competing models after data have been observed.

Discrete Random Variables

A **random variable** is a function that assigns a numerical value to each outcome of a random experiment. A **discrete random variable** takes values in a finite or countably infinite set.

Examples:

- X = number of heads in 3 coin flips, so $X \in \{0, 1, 2, 3\}$.
- Y = result of a die roll, so $Y \in \{1, 2, 3, 4, 5, 6\}$.
- N = number of emails received in an hour, so $N \in \{0, 1, 2, \dots\}$.

For a discrete random variable X , the **probability mass function** or **PMF** is:

$$p(x) = P(X = x)$$

The PMF must satisfy:

$$p(x) \geq 0 \quad \text{and} \quad \sum_x p(x) = 1$$

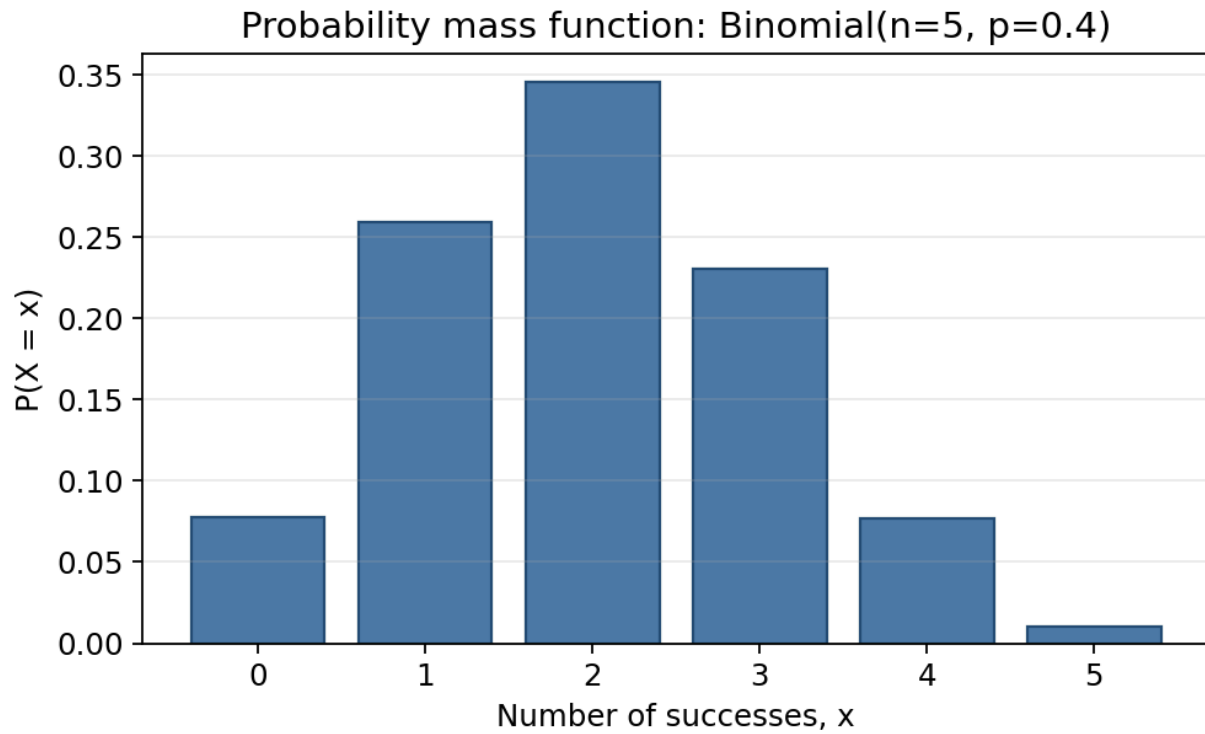


Fig. 4 A PMF assigns probability to each possible value of a discrete random variable.

For discrete random variables, conditional probability is given by:

$$P(X = x \mid Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} \quad \text{if } P(Y = y) > 0$$

Example: roll two fair dice. Let:

- X = value of the first die.
- S = sum of both dice.

What is $P(X = 4 \mid S = 7)$?

The outcomes with sum 7 are:

$$(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$$

There are 6 equally likely outcomes under the condition $S = 7$. Only one has $X = 4$, namely $(4, 3)$. Therefore:

$$P(X = 4 \mid S = 7) = \frac{1}{6}$$

Now ask a different question: what is $P(X \geq 4 \mid S = 7)$?

The favorable outcomes are:

$$(4, 3), (5, 2), (6, 1)$$

So:

$$P(X \geq 4 \mid S = 7) = \frac{3}{6} = \frac{1}{2}$$

Common Types of Discrete Random Variables

Several discrete distributions appear so often that they are worth recognizing.

Distribution	Values	Typical use	PMF
Bernoulli(p)	0, 1	One yes/no trial	$P(X = 1) = p,$ $P(X = 0) = 1 - p$
Binomial(n, p)	0, 1, $\dots$, n	Number of successes in n independent Bernoulli trials	$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
Geometric(p)	1, 2, $\dots$	Trial number of the first success	$P(X = k) = (1 - p)^{k-1} p$
Poisson(λ)	0, 1, 2, $\dots$	Counts in a fixed time or space interval	$P(X = k) = e^{-\lambda} \lambda^k / k!$
Discrete uniform	finite set	Equally likely outcomes	$P(X = x) = 1/n$ for n outcomes

The distribution you choose should reflect the data-generating process. For example, a Binomial model assumes a fixed number of trials, two outcomes per trial, constant success probability, and independence.

Example 1: Defective Parts

A factory produces parts. Each part is defective with probability $p = 0.02$, independently of the others. If we inspect $n = 10$ parts, let:

$$X = \text{number of defective parts}$$

Then:

$$X \sim \text{Binomial}(10, 0.02)$$

What is the probability of finding exactly one defective part?

$$P(X = 1) = \binom{10}{1} (0.02)^1 (0.98)^9$$

$$P(X = 1) = 10(0.02)(0.98)^9 \approx 0.1667$$

So there is about a 16.7% chance of exactly one defective part.

What is the probability of at least one defective part?

It is easier to use the complement:

$$P(X \geq 1) = 1 - P(X = 0)$$

$$P(X \geq 1) = 1 - \binom{10}{0} (0.02)^0 (0.98)^{10}$$

$$P(X \geq 1) = 1 - 0.98^{10} \approx 0.1829$$

So there is about an 18.3% chance of at least one defective part.

Example 2: Help Desk Tickets

Suppose the number of support tickets arriving in an hour is modeled as:

$$X \sim \text{Poisson}(3)$$

Here $\lambda = 3$ means the expected number of tickets per hour is 3.

What is the probability of exactly 5 tickets?

$$P(X = 5) = e^{-3} \frac{3^5}{5!}$$

$$P(X = 5) = e^{-3} \frac{243}{120} \approx 0.1008$$

What is the probability of at most 2 tickets?

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$P(X \leq 2) = e^{-3} \frac{3^0}{0!} + e^{-3} \frac{3^1}{1!} + e^{-3} \frac{3^2}{2!}$$

$$P(X \leq 2) = e^{-3}(1 + 3 + 4.5) \approx 0.4232$$

So the probability of at most 2 tickets in an hour is about 42.3%.

Independence and Conditional Independence

Independence describes when knowing one outcome provides no information about another. It is an assumption behind many probability models, including the independent Bernoulli trials used to construct a Binomial random variable.

Independence

Two events A and B are **independent** if

$$P(A \cap B) = P(A)P(B).$$

When the relevant conditional probabilities are defined, this is equivalent to

$$P(A | B) = P(A) \quad \text{and} \quad P(B | A) = P(B).$$

In words, learning that one event occurred does not change the probability of the other.

Two discrete random variables X and Y are independent, written

$$X \perp\!\!\!\perp Y,$$

if their joint PMF factors into the product of their individual PMFs for **every** pair of values x and y :

$$P(X = x, Y = y) = P(X = x)P(Y = y).$$

For example, let X and Y be the results of two independent rolls of a fair six-sided die. For any $x, y \in \{1, 2, 3, 4, 5, 6\}$,

$$P(X = x, Y = y) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = P(X = x)P(Y = y).$$

Knowing the result of the first roll therefore does not change the distribution of the second roll.

! Independence is not mutual exclusivity

Mutually exclusive events cannot occur together. If two mutually exclusive events both have positive probability, then they are **not** independent: observing one tells us that the other did not occur. Independence instead means that observing one event does not change the probability of the other.

Conditional Independence

Sometimes two random variables are dependent overall but become independent once a third variable is known. Discrete random variables X and Y are **conditionally independent given Z** , written

$$X \perp\!\!\!\perp Y \mid Z,$$

if, for every x , y , and z with $P(Z = z) > 0$,

$$P(X = x, Y = y \mid Z = z) = P(X = x \mid Z = z)P(Y = y \mid Z = z).$$

Under conditional independence, once Z is known, learning X provides no additional information about Y , and vice versa. This does **not** necessarily mean that X and Y are independent when Z is unknown.

Example: Two UAV Sensors with a Common Cause

Let three binary random variables represent an obstacle-detection problem:

- $O = 1$ if an obstacle is present and $O = 0$ otherwise.
- $C = 1$ if the camera raises an alert.
- $L = 1$ if the lidar raises an alert.

Suppose the camera and lidar have separate measurement noise. **Once we know whether an obstacle is present, assume their alerts are independent**, with

Condition	Camera alert probability	Lidar alert probability
Obstacle present ($O = 1$)	$P(C = 1 \mid O = 1) = 0.90$	$P(L = 1 \mid O = 1) = 0.80$
No obstacle ($O = 0$)	$P(C = 1 \mid O = 0) = 0.05$	$P(L = 1 \mid O = 0) = 0.10$

Conditional independence allows us to multiply within each condition:

$$P(C = 1, L = 1 \mid O = 1) = (0.90)(0.80) = 0.72,$$

and

$$P(C = 1, L = 1 \mid O = 0) = (0.05)(0.10) = 0.005.$$

Now suppose $P(O = 1) = 0.10$. Using the law of total probability, without knowing O , the marginal alert probabilities are

$$P(C = 1) = (0.10)(0.90) + (0.90)(0.05) = 0.135,$$

$$P(L = 1) = (0.10)(0.80) + (0.90)(0.10) = 0.17,$$

while the probability that both sensors alert is

$$P(C = 1, L = 1) = (0.10)(0.72) + (0.90)(0.005) = 0.0765.$$

Because

$$0.0765 \neq (0.135)(0.17) = 0.02295,$$

C and L are not independent when the obstacle state is unknown. They are associated through their common cause, O , but they are conditionally independent once O is known.

Expectation and Variance

The **expectation** of a discrete random variable is its probability-weighted average:

$$E[X] = \sum_x xP(X = x)$$

Expectation is not necessarily a value that X can actually take. A fair die has:

$$E[X] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \cdots + 6 \cdot \frac{1}{6} = 3.5$$

No die roll is 3.5, but 3.5 is the long-run average roll.

The **variance** measures spread around the expectation:

$$\text{Var}(X) = E[(X - E[X])^2]$$

For a discrete random variable:

$$\text{Var}(X) = \sum_x (x - \mu)^2 P(X = x) \quad \text{where } \mu = E[X]$$

A useful equivalent formula is:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

where:

$$E[X^2] = \sum_x x^2 P(X = x)$$

For common distributions:

Distribution	$E[X]$	$\text{Var}(X)$
Bernoulli(p)	p	$p(1-p)$
Binomial(n, p)	np	$np(1-p)$
Geometric(p)	$1/p$	$(1-p)/p^2$
Poisson(λ)	λ	λ

Law of Total Expectation

Sometimes the distribution of X depends on another random variable Z . For example, the number of requests received by a server may depend on whether the server is under normal or heavy load. If we know the conditional expectation of X within each possible state of Z , we can recover the overall expectation by averaging those conditional expectations:

$$E[X] = E_Z[E[X | Z]].$$

When Z is discrete, this becomes

$$E[X] = \sum_z E[X | Z = z]P(Z = z).$$

This result is called the **law of total expectation**, or sometimes the **tower property**. It says that we may first compute an average within each group and then average those group means according to how often the groups occur.

More generally, for any function g ,

$$E[g(X)] = E_Z[E[g(X) | Z]].$$

Law of Total Variance

There is a corresponding result for variance, but simply averaging the conditional variances is not enough. Overall variability can arise from two sources:

1. variation of X **within** each state of Z , and
2. variation of the conditional means **between** the states of Z .

The **law of total variance** combines both sources:

$$\text{Var}(X) = E_Z[\text{Var}(X | Z)] + \text{Var}_Z(E[X | Z]).$$

The first term is the average within-group variance. The second term is the variance of the group means. It is zero only when all states of Z have the same conditional expectation.

The formula follows from

$$E[X^2 | Z] = \text{Var}(X | Z) + (E[X | Z])^2.$$

Taking expectations and subtracting $(E[X])^2$ gives the two terms in the law of total variance.

Example: Requests under changing server load

Suppose Z records the server load:

$$P(Z = \text{normal}) = 0.8, \quad P(Z = \text{busy}) = 0.2.$$

Let X be the number of requests received in one minute, with

$$X \mid Z = \text{normal} \sim \text{Poisson}(2)$$

and

$$X \mid Z = \text{busy} \sim \text{Poisson}(6).$$

Because the expectation of a Poisson random variable equals its rate, the law of total expectation gives

$$\begin{aligned} E[X] &= (0.8)(2) + (0.2)(6) \\ &= 2.8. \end{aligned}$$

The conditional variances are also 2 and 6. Therefore, the average within-mode variance is

$$E[\text{Var}(X \mid Z)] = (0.8)(2) + (0.2)(6) = 2.8.$$

The variance between the two conditional means is

$$\begin{aligned} \text{Var}(E[X \mid Z]) &= (0.8)(2 - 2.8)^2 + (0.2)(6 - 2.8)^2 \\ &= 2.56. \end{aligned}$$

Thus the overall variance is

$$\text{Var}(X) = 2.8 + 2.56 = 5.36.$$

The marginal variance is larger than the average conditional variance because the changing server mode introduces additional variation in the mean request rate.

Probability and Likelihood

Probability and likelihood use the same mathematical model but answer different questions. Suppose a PMF is written as $p(x \mid \theta)$, where x is a possible observation and θ is a model parameter for a given distribution.

Concept	Held fixed	Allowed to vary	Question
Probability $p(x \mid \theta)$	Model parameter θ	Possible data x	If this model is fixed, how probable is each possible outcome?
Likelihood $L(\theta; x) = p(x \mid \theta)$	Observed data x	Candidate parameter θ	Given the data we observed, which parameter values are better supported?

As a probability distribution over x , the PMF must satisfy

$$\sum_x p(x \mid \theta) = 1$$

for each fixed value of θ . A likelihood function is viewed as a function of θ and does not generally sum or integrate to 1 over the possible parameter values.

For example, suppose X is the number of successful telemetry transmissions in 10 independent attempts, each with success probability p . Before observing X , if $p = 0.8$ is fixed, the probability of exactly 8 successes is

$$P(X = 8 \mid p = 0.8) = \binom{10}{8} (0.8)^8 (0.2)^2 \approx 0.302.$$

After observing $X = 8$, the same Binomial expression becomes a likelihood function for the unknown success probability:

$$L(p; X = 8) = \binom{10}{8} p^8 (1 - p)^2.$$

Now the observation 8 is fixed and different values of p are compared. The likelihood is largest at $p = 0.8$, which is the maximum likelihood estimate in this example.

Likelihood is not a posterior probability

$L(p; X = 8)$ is not $P(p \mid X = 8)$. To construct a probability distribution over the unknown parameter p , we must specify a prior distribution and apply Bayes' theorem. More on this later.

Summary

A discrete random variable maps outcomes to a finite or countably infinite set of numbers, and its PMF assigns probability to those values. Bernoulli, Binomial, Geometric, Poisson, and discrete uniform distributions model different data-generating processes. Probability treats the model as fixed and varies the possible data; likelihood treats the observed data as fixed and compares model parameters. Independence means that one variable supplies no information about another, while conditional independence means this holds after a third variable is known. Expectation and variance summarize the center and spread of a distribution, and the laws of total expectation and total variance combine conditional moments across different states of another variable.

Practice Questions and Solutions

1. Discrete Random Variable: Telemetry Packets

A UAV transmits five telemetry packets during a maneuver. Each packet is received successfully with probability 0.80, independently of the other packets. Let X be the number of packets received successfully.

1. What distribution does X follow?
2. What is the probability that at least four packets are received successfully?
3. Find $E[X]$ and $\text{Var}(X)$.

Solution

Click to show 

2. Discrete Random Variable: Wind Gusts

The number of unexpected strong wind gusts encountered during a UAV flight is modeled as a Poisson random variable with an average of two gusts per flight. Let Y be the number of gusts.

1. What is the probability that the UAV encounters at most one gust?
2. What are the expected value and variance of Y ?

 Solution

Click to show 

3. Mean and Variance: GPS Lock Acquisition

Before beginning a mission, a UAV repeatedly scans for a reliable GPS lock. Each scan succeeds with probability 0.75, independently of previous scans. Let Z be the number of scans required to obtain the first reliable GPS lock, including the successful scan.

1. What distribution does Z follow?
2. Find $E[Z]$ and $\text{Var}(Z)$.
3. Briefly interpret the expected value in the context of the UAV mission.

 Solution

Click to show 

Reading

Christopher M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006, select parts of Appendix B.

References

- Joseph K. Blitzstein and Jessica Hwang, *Introduction to Probability*, 2nd ed., CRC Press, 2019. Book site: <https://projects.iq.harvard.edu/stat110/home>
- OpenStax, *Introductory Statistics*, probability and discrete random variable chapters: <https://openstax.org/details/books/introductory-statistics>
- An exceptional set of lectures on probability and statistics by Steven Brunton of UWash: <https://youtube.com/playlist?list=PLMrJAKhleNNR3sNYvfqiKqcStwuPSts9V&si=sFO1ISGIWPbBZHqg/>
- Companion code notebook for plotting the discrete probability distributions: [discrete probability plotting notebook](#)